ANSWER KEY & MARKING SCHEME SET A

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** a prime number	definition
2	**(d)** 51	51 = 3 × 17
3	**(a)** Sieve of Eratosthenes	ancient method
4	**(c)** 0 or 5	rule
5	**(a)** 2	first prime
6	**(c)** 1 and itself	basic fact
7	**(c)** 2, 3, 5	30 = 2×3×5
8	**(d)** less than or equal to the number	property
9	**(a)** 7	common factors of 14 and 21
10	**(a)** 5	common factors of 10 and 35

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A number with exactly two factors, 1 and itself; 2, 3, 5.	1+1
12	1, 2, 3, 4, 6, 8, 12, 24.	1+1
13	7, 14, 21, 28, 35.	1+1
14	6.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	84 = 2×2×3×7; 126 = 2×3×3×7 (2); HCF = 42 (1).	2+1
16	Write numbers 1 to 100 (1); cross out 1 and multiples of 2, 3, 5, 7 keeping the prime (1); remaining numbers are primes (1).	1+1+1
17	12 = $2^2 \times 3$; 18 = 2×3^2 (1.5); LCM = $2^2 \times 3^2$ = 36 (1.5).	1½+1½
	OR: Divisible by 2, 3 (digit sum 18), 4 (last two digits 56), 6 and 9 (3).	1+2
18	2, 3, 5, 7, 11, 13, 17, 19, 23, 29 (2); 10 primes (1).	2+1
	OR: Any common factor of n and n+1 divides their difference 1 (1.5); e.g. 8 and 9 are co-prime (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2^2 \times 3^2 \times 5$ (2). (ii) 18 (1.5). (iii) 1, 2, 3, 4, 5, 6, 9, 10, 12, 15, 18, 20, 30, 36, 45, 60, 90, 180 (1.5).	2+1½+1½
	OR: (i) 24 = $2^3 \times 3$, 36 = $2^2 \times 3^2$; HCF = 12, LCM = 72 (3). (ii) 12 × 72 = 864 = 24 × 36 (2).	3+2
20	(i) Rules (3). (ii) Divisible by 2, 3, 4, 5, 6, 9, 10 (2).	3+2
	OR: (i) HCF = 12 kits (1). (ii) 3 (1). (iii) 4 (1). (iv) HCF (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** neither prime nor composite	only one factor
2	**(b)** co-prime	e.g. 8 and 15
3	**(b)** 0, 2, 4, 6 or 8	even
4	**(d)** 0	rule
5	**(b)** 8	2,3,5,7,11,13,17,19
6	**(d)** highest common factor	definition
7	**(d)** 11	first prime ≥ 10
8	**(c)** 6	common factors of 12 and 18
9	**(d)** 8	common factors of 16 and 24
10	**(a)** 12	smallest common multiple of 4 and 6

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	20.	1+1
12	Numbers with common factor only 1; 9 and 10.	1+1
13	No; 91 = 7 × 13.	1+1
14	$2 \times 2 \times 3 \times 3$.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	HCF of 48 and 36 = 12 (1); the pieces are 12 m (2).	1+2
16	LCM of 6, 8, 12 = 24 minutes (3).	1+2
17	1, 2, 3, 4, 6, 9, 12, 18, 36 (2); 9 factors (1).	2+1
	OR: 2 × 3 × 3 × 5 (2); four prime factors (1).	2+1
18	HCF is 1 (2); so co-prime (1).	2+1
	OR: Digit sum divisible by 9 (1); 7+1+2+8 = 18, so divisible (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47 (2). (ii) 15 (1). (iii) (3,5), (5,7), (11,13), (17,19), (29,31), (41,43) (2).	2+1+2
	OR: (i) HCF 1 (1). (ii) (21,22) and (35,36) (2). (iii) Gear teeth / safe locks (2).	1+2+2
20	(i) LCM = 120 s (1). (ii) 5 times after the start (1). (iii) LCM (1). (iv) 63 s (2).	1+1+1+2
	OR: (i) Process (3). (ii) 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.